

2 Different techniques can be used in the investigation of the activity of the brain.

(a) Which type of scan can be used to study changes in brain function as they happen?

(1)

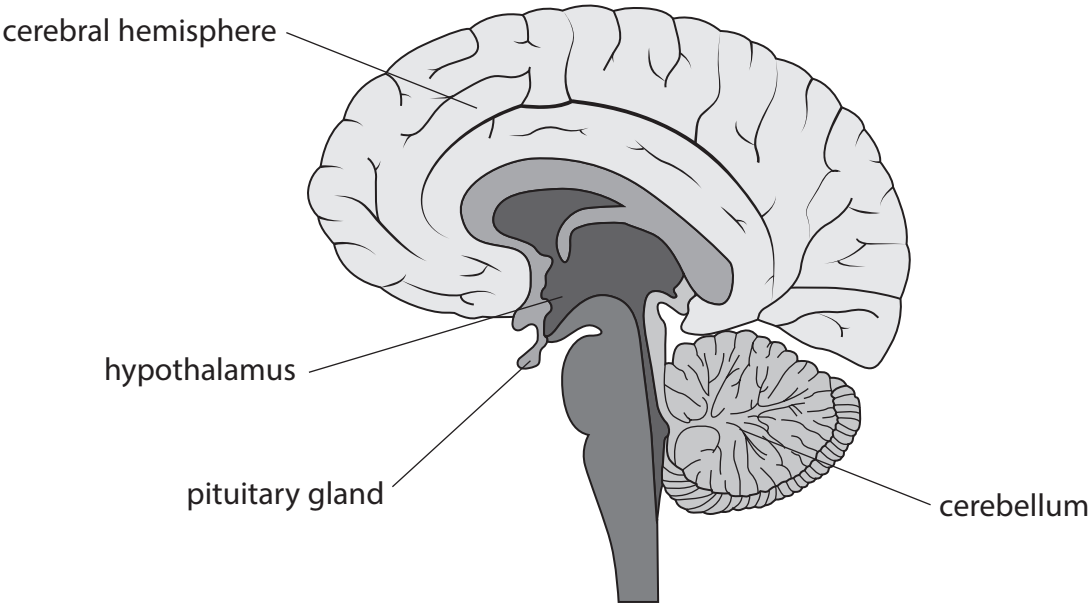
- ☐ **A** computed tomography (CT)
- ☐ **B** electrocardiogram (ECG)
- ☐ **C** functional magnetic resonance imaging (fMRI)
- ☐ **D** polymerase chain reaction (PCR)

(b) Describe the advantages and disadvantages of using positron emission tomography (PET) and magnetic resonance imaging (MRI) to investigate brain function.

(3)

[illegible]

(c) The diagram shows a section through the human brain.



Complete the table to show **one** function for each part of the brain.

(3)

Part of brain	Function
cerebellum	
cerebral hemisphere	
hypothalamus	

(Total for Question 2 = 7 marks)